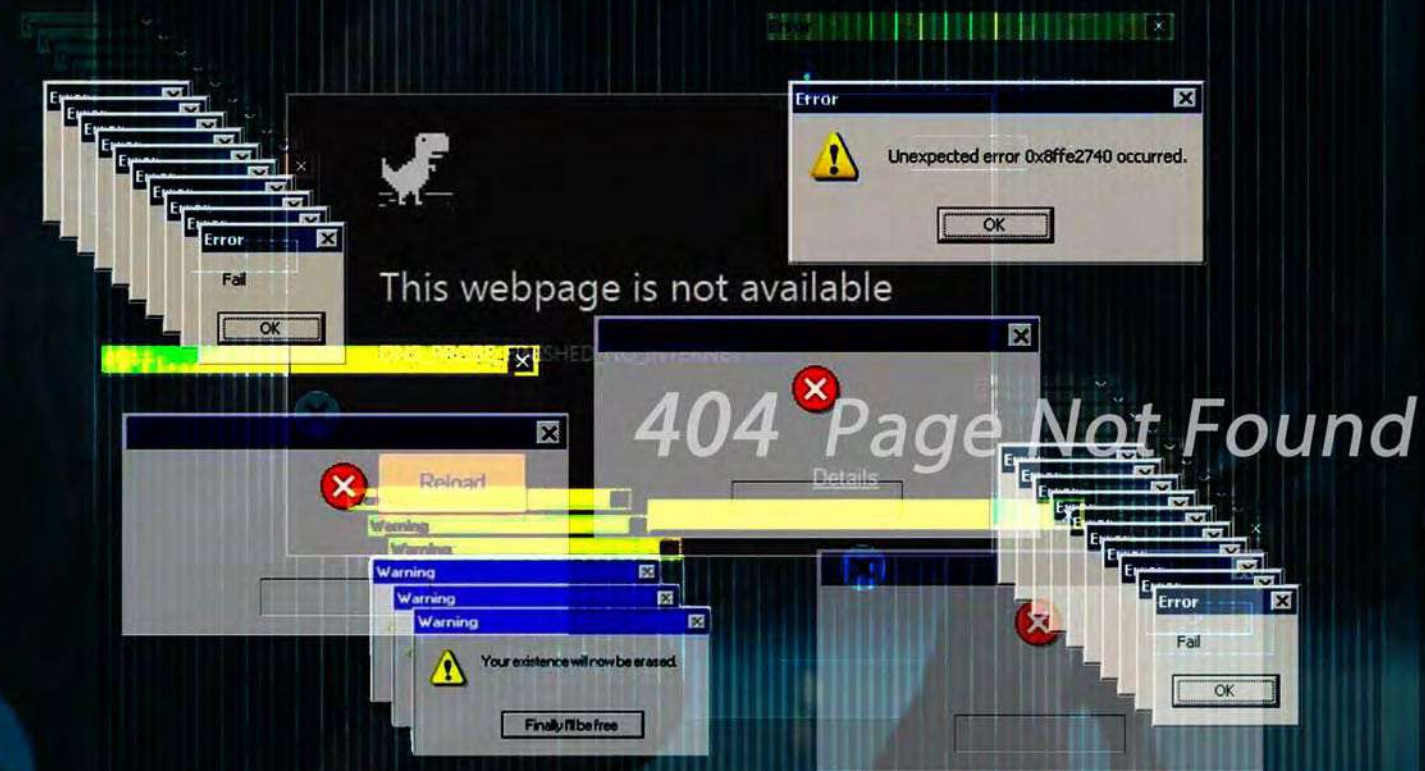
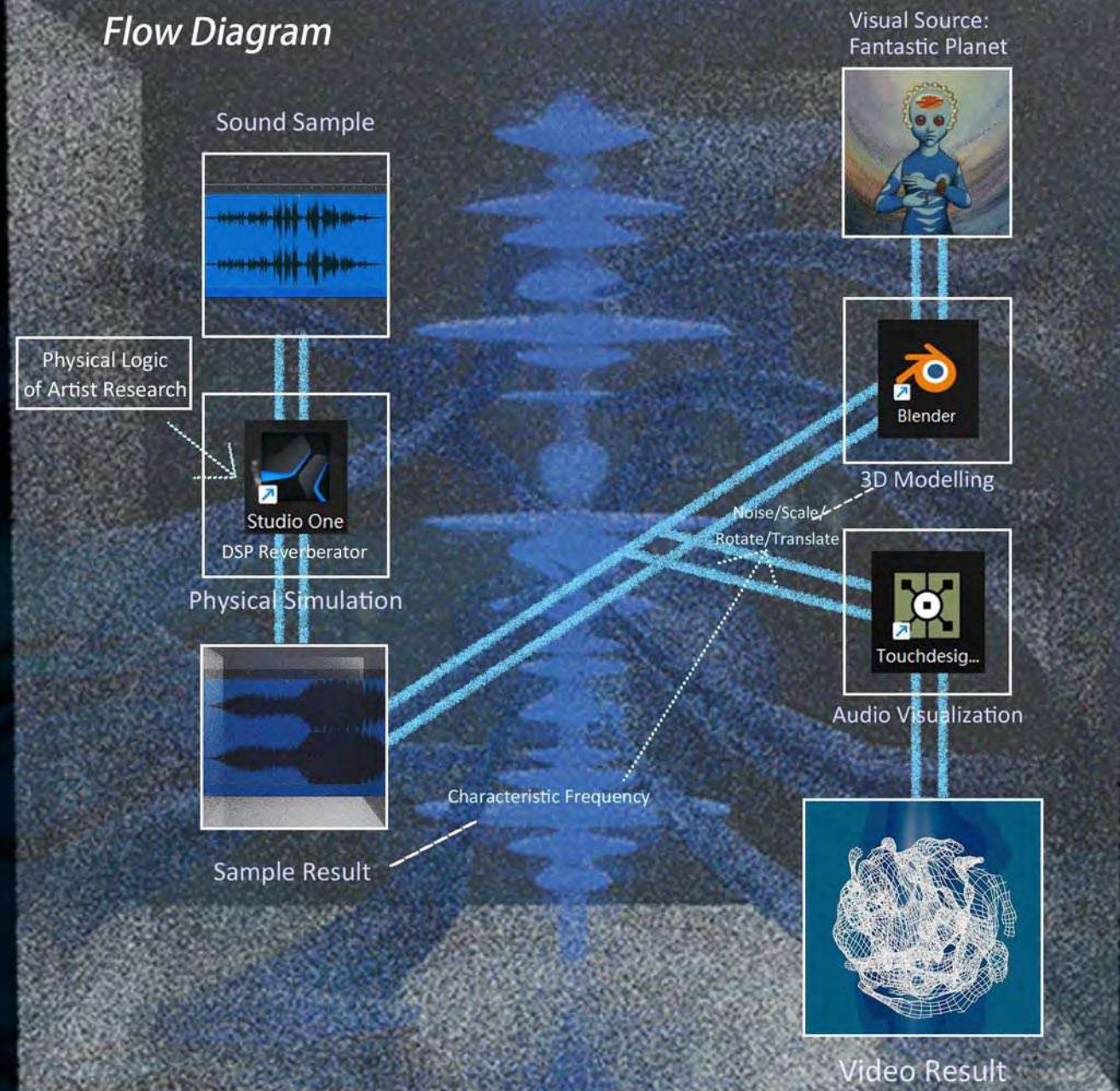




The Walls of China

Flow Diagram





Designer: Dominic Duan
Photographer: Tang
Model: Jiang

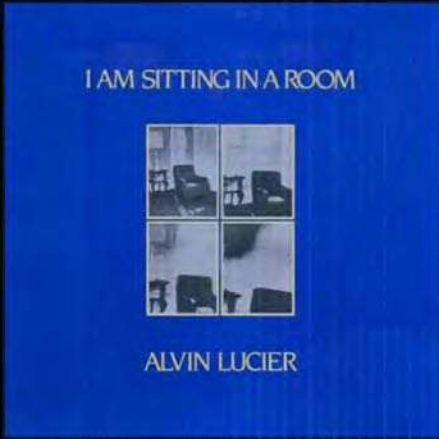






Designer: Dominic Duan
Photographer: Tang
Model: Jiang





Sound Experiment

<https://www.youtube.com/watch?v=zjiX4oc8e2U&list=WL&index=27&t>

I researched Alvin Lucier's 1969 sound experiment 'I Am Sitting in a Room.' Its physical logic inspired me: the human voice serves as the excitation source and the room as the excited system. In a closed room, sound repeats, amplifying the room's characteristic frequencies and absorbing others. Over time, only the room's characteristic frequencies remain.



Sound Sample Processing

I documented how the original audio's characteristic frequencies were gradually replaced by the system's frequencies.

Tools:
Studio One, Valhalla Reverb

1. Sample
2. Plugin: Reverb
3. Mixed down to a new track
4. Repeat 10 times



Initial characteristic frequencies



Final characteristic frequencies



Inspired by the movie "Fantastic Planet", which tells the story of the Oms who, after being enslaved and treated as toys by the Draags, learn advanced knowledge and rebel against their rulers, I created 3D models based on the wild Oms' life hidden in tree hollows during their oppression. I focused on their environment and colors to depict a society under strict control and limited information.

Process



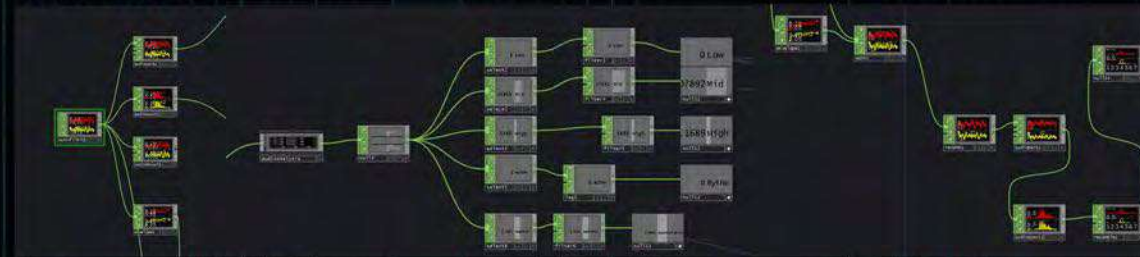
Three-view



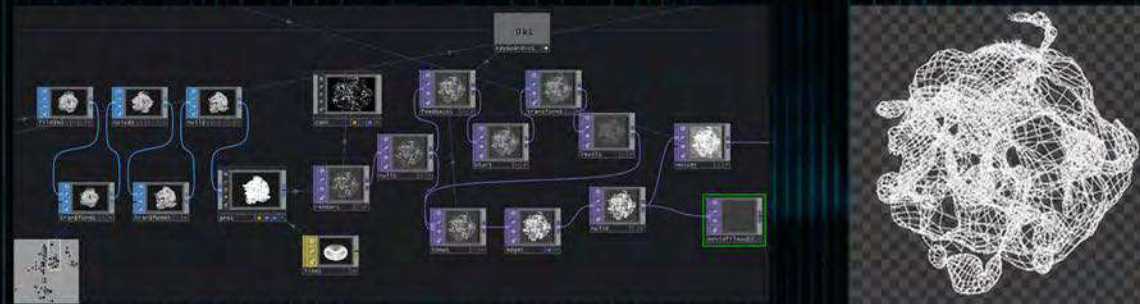
Then I used the audio from the sound experiment, utilizing its frequency, speed, and other parameters to control the morphological changes of the 3D model. This audio visualization illustrates the societal conditions under control.



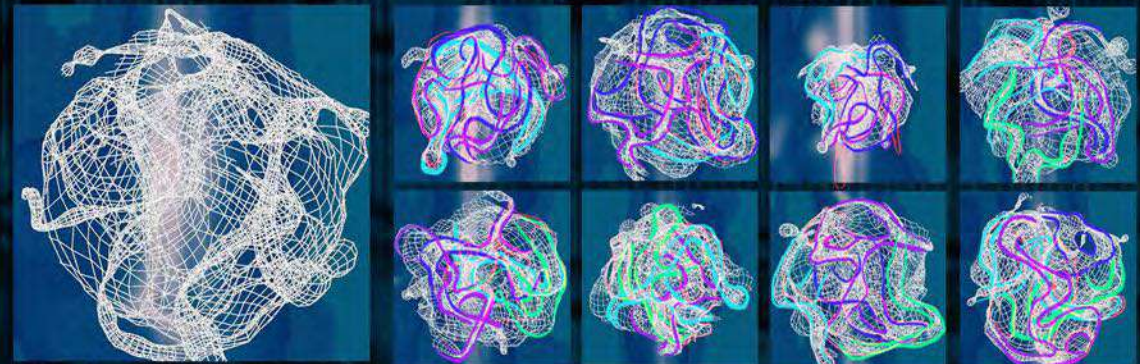
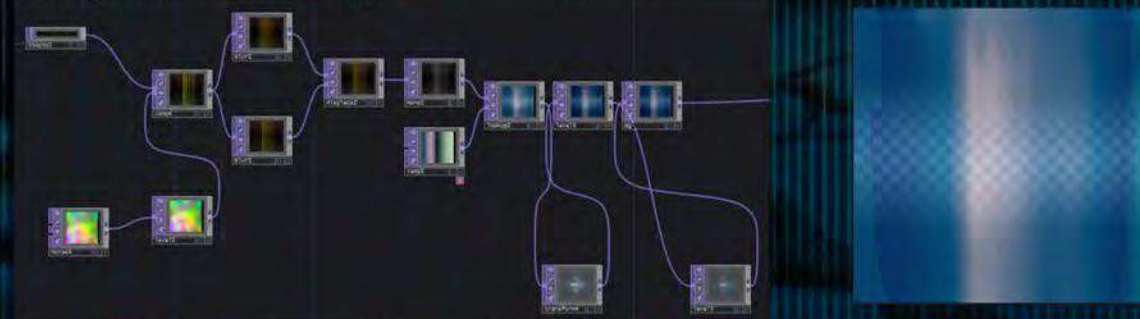
Audio Analysis



3D Model Processing



Background Processing



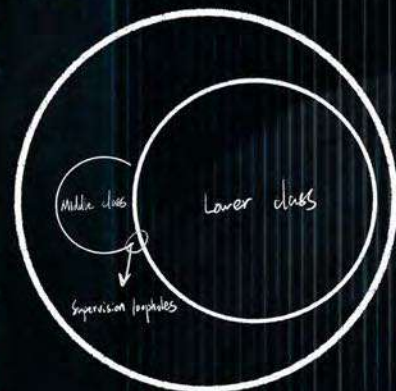
https://youtu.be/EBxTM3i_cPM



Pattern Experiment

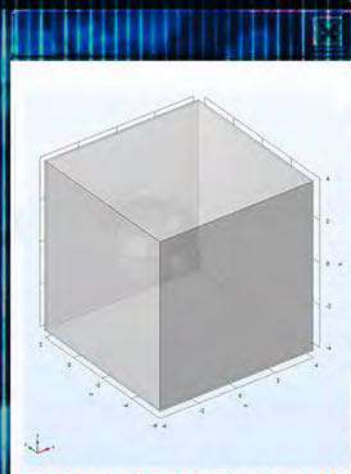
File Edit View Help

This sketch represents a society where the middle class is smaller than the lower class, and speech is heavily censored. Most voices are suppressed, with the lower class almost entirely silenced. However, some voices from the middle class slip through supervision loopholes, offering effective solutions.

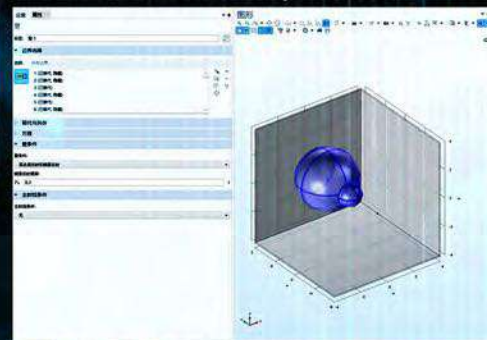


I modeled this concept and used COMSOL to simulate sound ray trajectories, capturing the leaked voices.

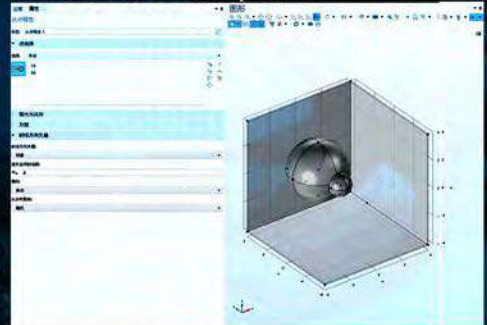
1. Modeling



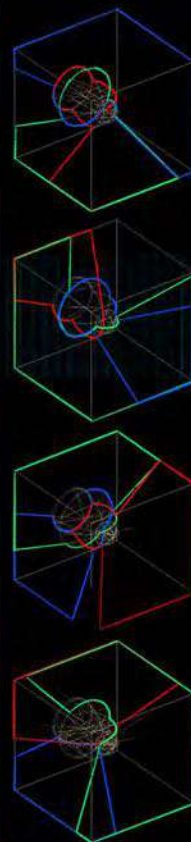
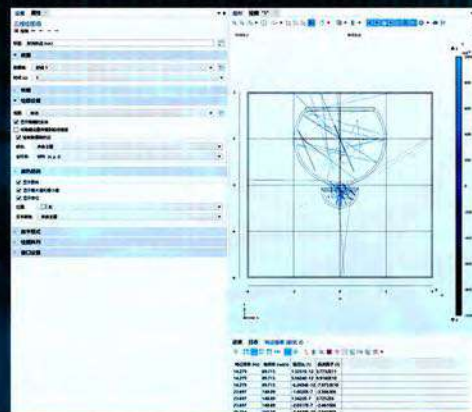
2. Set reflective surface parameters



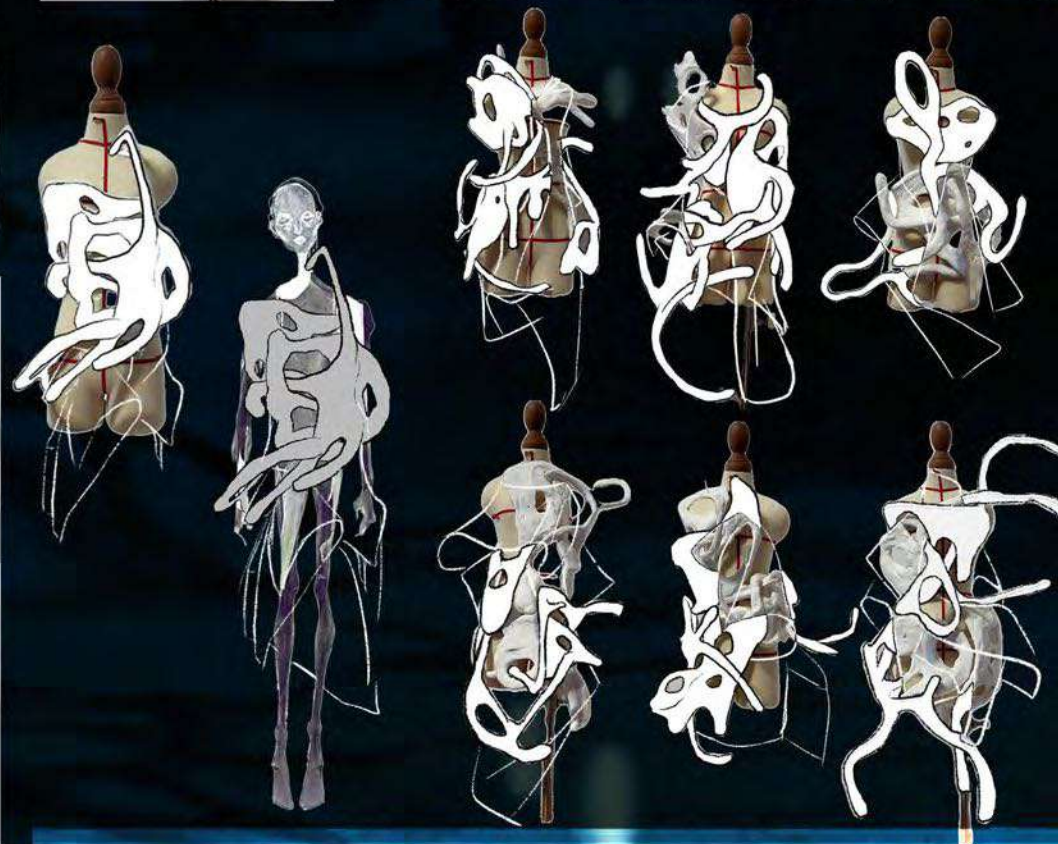
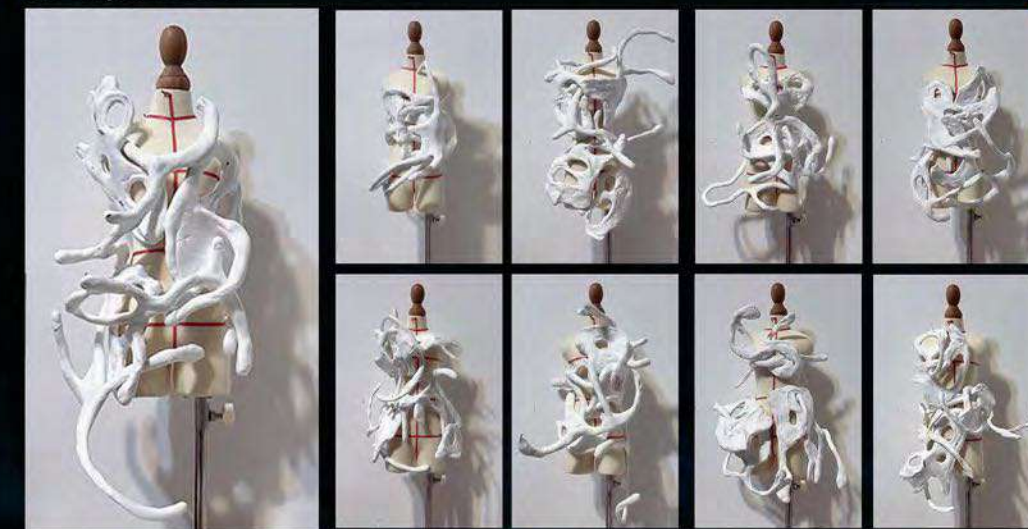
3. Set sound source points



4. Calculate ray trajectories



3D Experiment

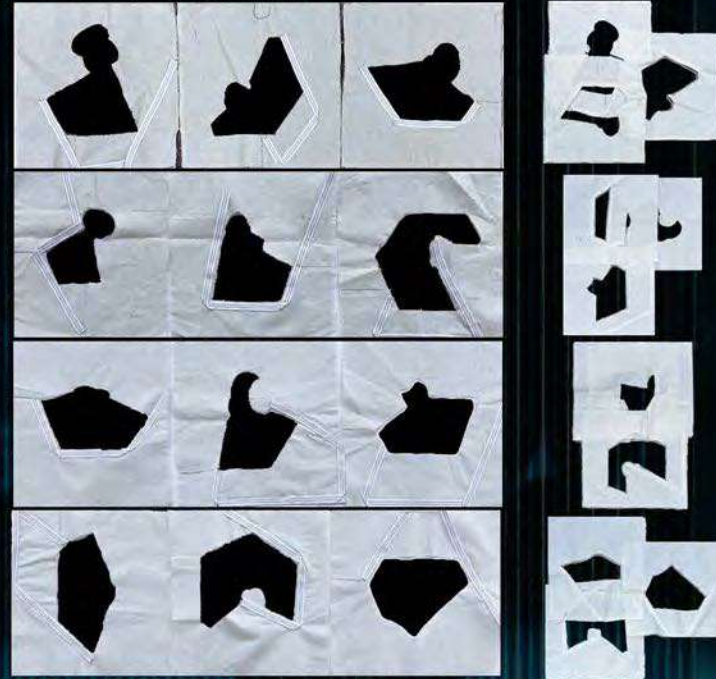


3D Development

Integrate the pattern pieces with the 3D elements



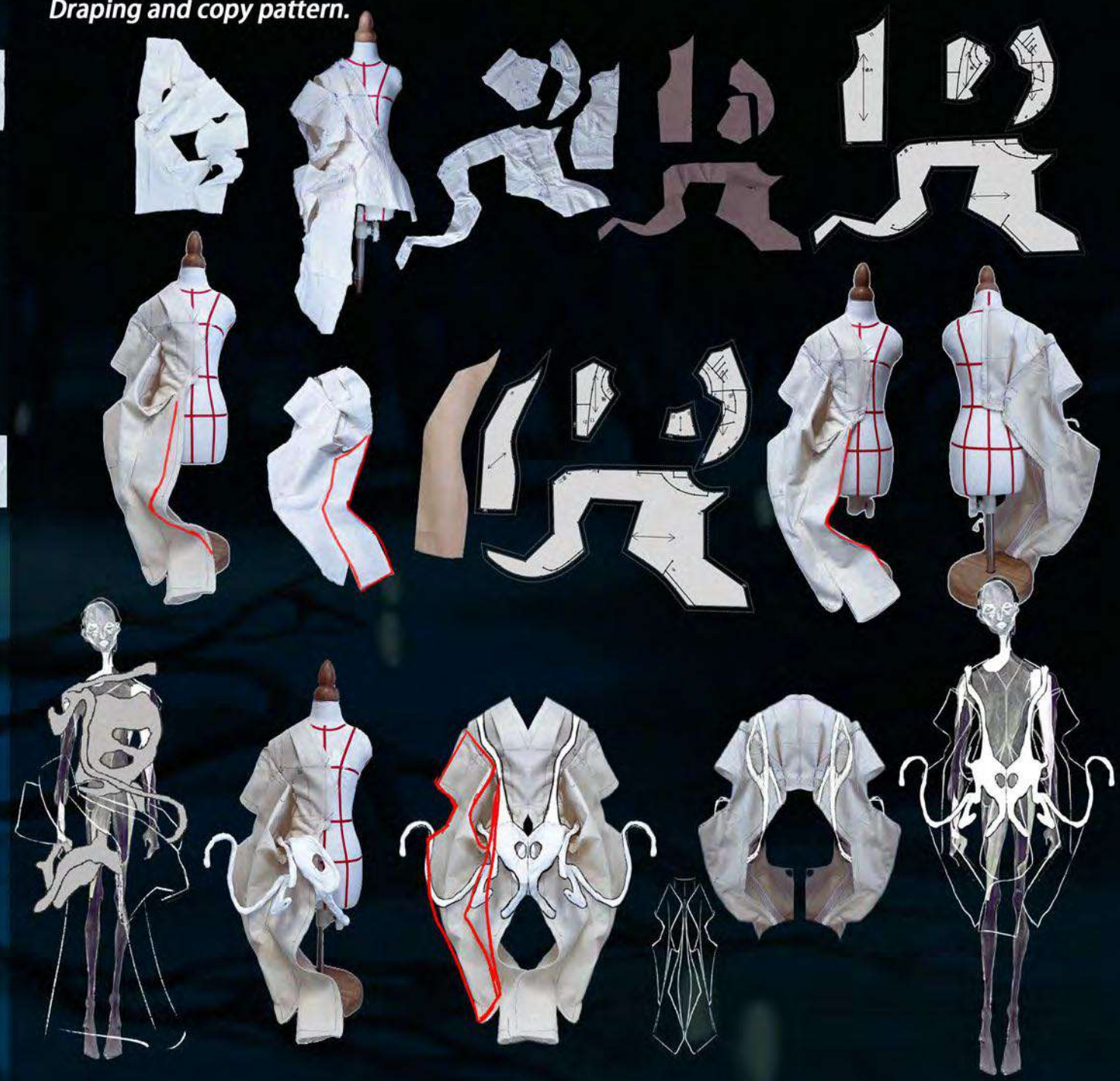
Pattern Deepening



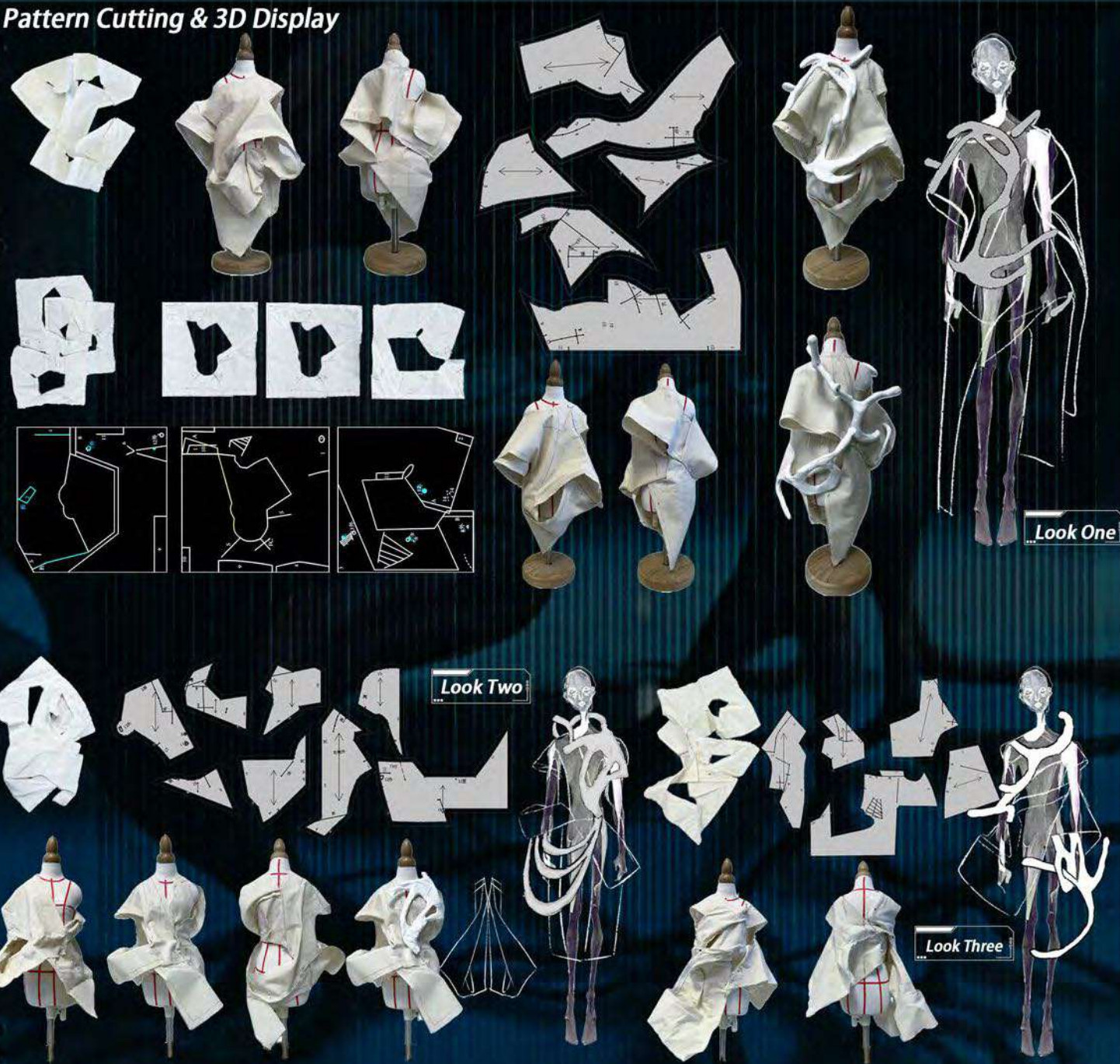
1. Applying boning along the trajectory derived from the physics experiment.
2. Combine a set of patterns into one.



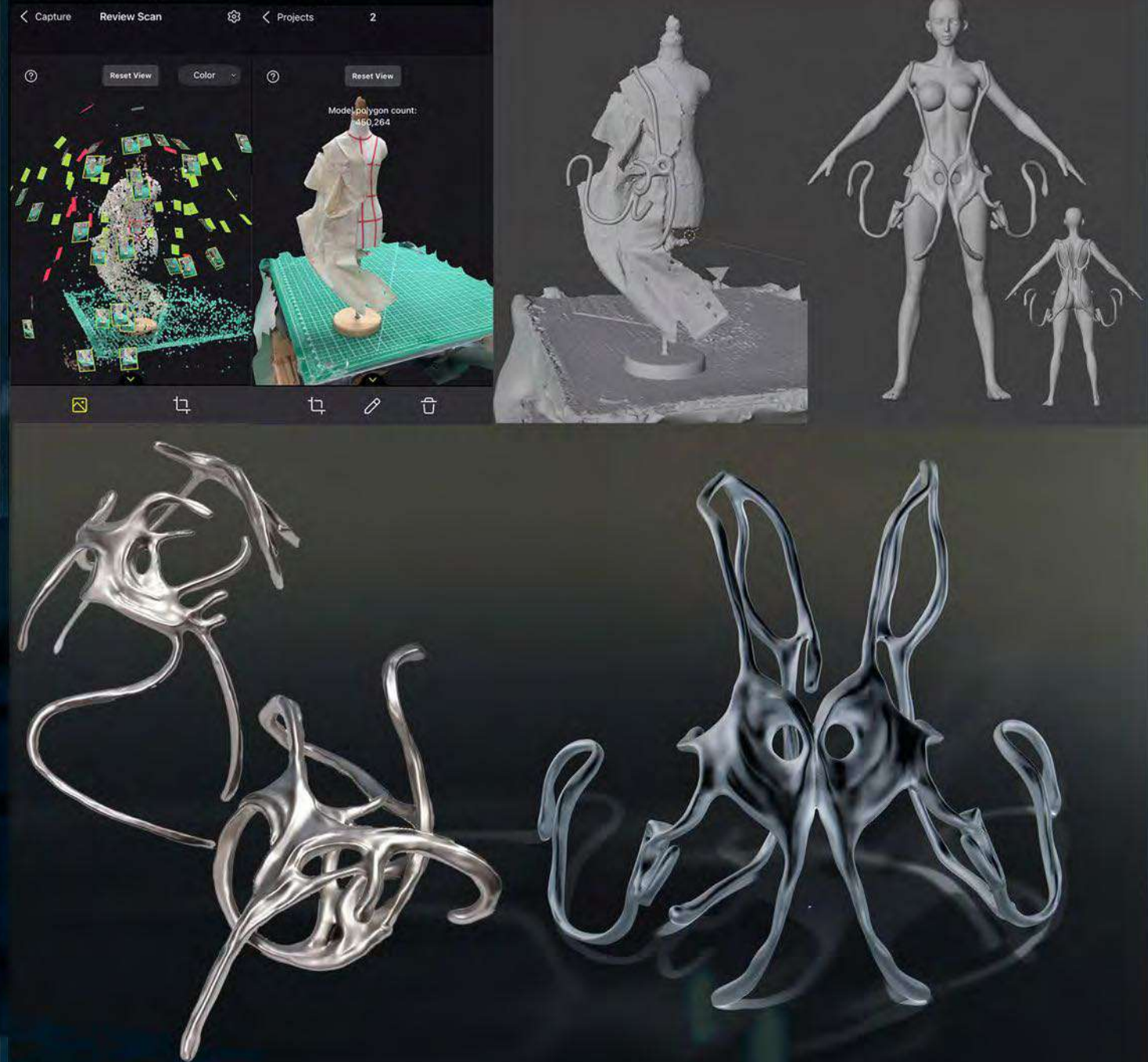
Draping and copy pattern.



Pattern Cutting & 3D Display



3D Scanning and Modeling



To get through the 'wall', I want to express how information flows on fabric and write code to simulate the process of information processing.

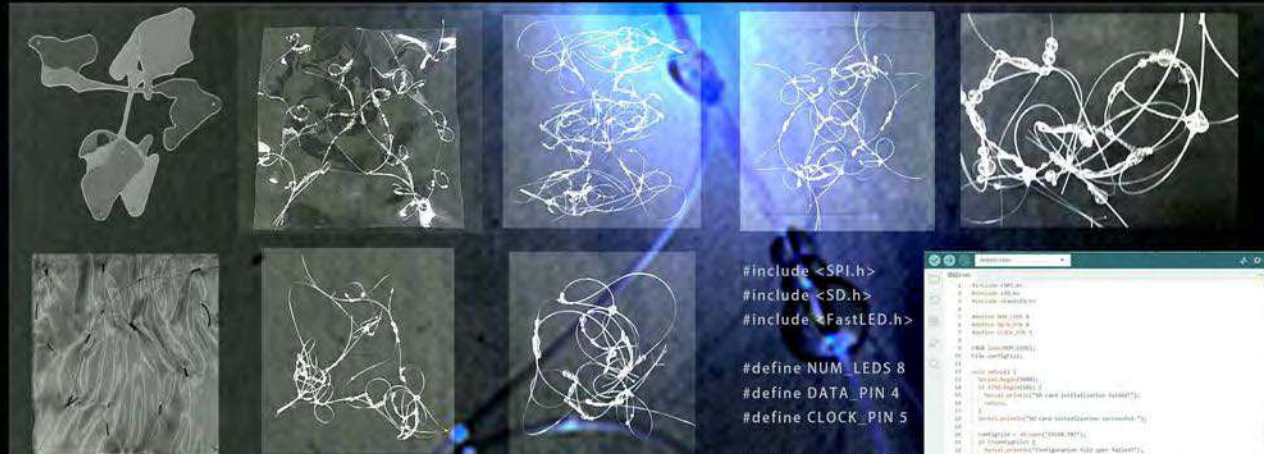
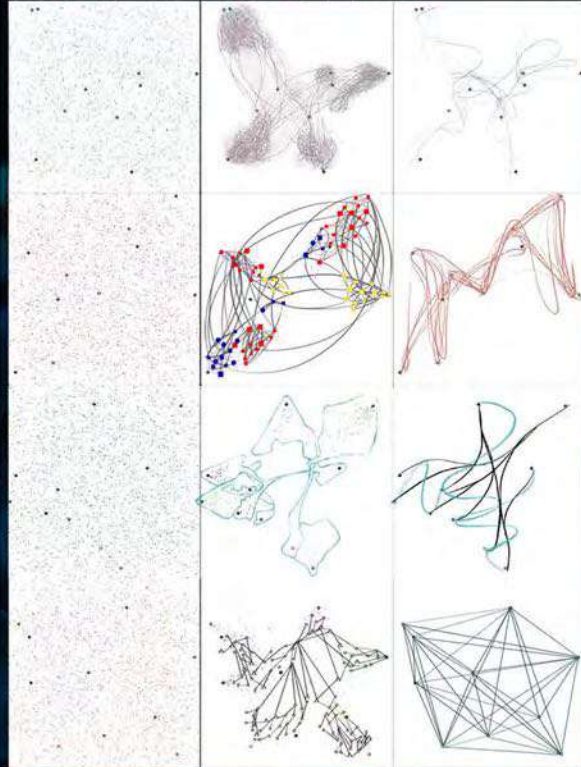
Figure 28 SYMAP of Cypso-Geometric to Hellenistic components in the Khrysokhou drainage.

Key: - Zone I
 +, O, + Intermediate Zone I/II
 II Zone II
 h, f Single component on single site locus
 S Multiple components on single site locus



The second approach simulates the process of clustering and analyzing noise information.

The screenshot shows the Processing IDE interface. At the top, the menu bar includes '文件' (File), '编辑' (Edit), '运行' (Run), '工具' (Tools), '帮助' (Help), 'sketch_聚类 | P...', and a window icon. Below the menu bar, there are two circular buttons on the left (one with a play icon, one with a square icon) and a circular button on the right with the text '运行' (Run). The main area shows the 'sketch' menu open, displaying a list of items: 'run numPoints = 5000; // 增加点的数量', 'PVector[] points = new PVector[numPoints];', 'color[] colors = new color[numPoints];', 'int numClusters = 10; // 增加聚类数', 'color[] centroids = new color[numClusters]; // 使用颜色作为质心', 'PVector[] centroidPositions = new PVector[numClusters];', 'color[] clusterColors = new color[numClusters];', 'int[] clusterAssignments = new int[numPoints];', 'int frameCount = 0;', 'void setup() {', 'size(800, 800);', 'generateData();', 'initClusters();', '}', and '}'. The 'run' button is highlighted in blue.



Lighting Interaction

Materials List: Optical fiber, Arduino UNO, SK9822 5V, SD Card Module



Processing:

1. Use Python to process the video, calculate the average color for 6-8 LEDs, and generate an RGBW value for each frame, saving it as a .txt file.
2. Save the .txt file to the SD module.
3. Arduino reads the color data from the .txt file to control the LEDs.

```
#include <SPI.h>
#include <SD.h>
#include <FastLED.h>
```

```
#define NUM_LEDS 8
#define DATA_PIN 4
#define CLOCK_PIN 5
```

```
CRGB leds[NUM_LEDS];
File configFile;
```

```
void setup() {  
  Serial.begin(9600);  
  if (!SD.begin(10)) {  
    Serial.println("SD card initialization failed!");  
    return;  
  }  
  Serial.println("SD card initialization successful.");  
}
```

```
configFile = SD.open("COLOR.TXT");
if (!configFile) {
  Serial.println("Configuration file open failed!");
  return;
}
Serial.println("Configuration file opened successfully.");
```

```
FastLED.addLeds<APA102, DATA_PIN, CLOCK_PIN, BRG>(leds, NUM_LEDS);
```

```
void loop() {
  while (configFile.available()) {
    String line = configFile.readStringUntil('\n');
    if (line.length() > 0) {
      int r, g, b, w;
      char *token = strtok((char *)line.c_str(), " ");
      int index = 0;
```

```
while (token != NULL && index < NUM_LEDS) {
    sscanf(token, "%d,%d,%d,%d", &r, &g, &b, &w);
```

```
// 选择降低白色通道的影响，可以乘以一个系数
w = w * 0.2; // 将白色通道减少50%
```

```
leds[index] = CRGB(r - w, g - w, b - w); // 减去白色通道值
```

```
token = strtok(NULL, " ");
```


Final Line-up



3D Clo & Fitting Process
- Look One



Look Three

